

If a student has scored less than 50, his status must be shown as 'FAIL'. Otherwise, his mark must be displayed.

The 3 developers wrote the following queries:

Tom:

```
SELECT STUDENTID, CASE WHEN MARKS < 50 THEN 'FAIL' ELSE TO_CHAR(Marks) END AS STATUS FROM ASSESSMENT
```

Dick:

```
SELECT STUDENTID, CASE WHEN MARKS < 50 THEN 'FAIL' ELSE Marks END AS STATUS FROM ASSESSMENT
```

Harry:

```
SELECT STUDENTID, CASE WHEN MARKS > 50 THEN 'FAIL' ELSE Marks END FROM ASSESSMENT
```

Which one of them will give the desired output?

- ☐ Dick
- ☐ Harry
- ☒ Tom
- ☐ None of them

Q5 of 20

A table Employee has the following data:

EmployeeNo	EmployeeName	Department	Salary
1111	Paul	Sales	1000
1222	Smith	Sales	200
2111	Gary	Marketing	1000
2222	Manya	Marketing	500
3111	Raj	HR	1000
3222	Paul	HR	200

The following queries were executed on the table successfully :

```
UPDATE EMPLOYEE SET DEPARTMENT = 'HR' WHERE DEPARTMENT = 'Marketing' ;
```

```
DELETE FROM EMPLOYEE WHERE DEPARTMENT = 'HR' AND SALARY = 1000;
```

What will be the output of the following query?

```
SELECT COUNT(*) FROM EMPLOYEE;
```

- ☐ 5
- ☐ 6
- ☒ 4
- ☐ 2

Q3 of 20



Tables course and student have 1-N relationship respectively. Cid is the primary key of course table and Sid is the primary key of student table. To which table the foreign key should be added?

Course
Cid
Cname

Student
Sid
Sname

- ☐ Either Student or Course
- ☐ In 1-N relationship foreign key cannot be established
- ☒ Only Student
- ☐ Only Course

Q2 of 20

Consider the following code written for creating the table:

```
CREATE TABLE ACCOUNT (ACCNO INT, ACCNAME VARCHAR(30) NOT NULL, BALANCE );
```

The table is NOT getting created, identify the reason.

- ☐ BALANCE must be NOT NULL
- ☐ ACCNO must be NOT NULL
- ☐ Primary key is missing for ACCOUNT
- ☒ BALANCE must have a datatype

Q6 of 20

Consider the following statements with respect to a candidate key:

- a. Candidate key identifies rows in a relation uniquely.
- b. There can be only one candidate key in a relation.
- c. A candidate key can be a combination of more than one attribute in a relation.

Identify the statement(s) which are TRUE.

- ☒ Only a and c
- ☐ Only a and b
- ☐ only a
- ☐ Only b and c

Q4 of 20

Rajesh created a table EMP in order to record employee details.

The table creation script for the same is given below:

```
CREATE TABLE EMP(  
empid NUMBER(10) PRIMARY KEY,  
empname VARCHAR2(50),  
cabinnumber NUMBER(20) UNIQUE  
);
```

Currently table has some data as given below:

empid	empname	cabinnumber
1001	Hari	789
1002	Varun	145
1003	Shyam	458

Select the right option for inserting a new row into the table.

- ☐ INSERT INTO EMP VALUES (1004,'Lali',789);
- ☒ INSERT INTO EMP VALUES (1004,'Lali',NULL);
- ☐ INSERT INTO EMP VALUES (NULL,'Lali',456);
- ☐ INSERT INTO EMP VALUES (1003,'Lali',578);

Q7 of 20

Consider the below table SalesPerson:

Id	Amount
1002	2400
1005	1800
1001	460
1002	2400
1009	600
1009	600
1009	600

SELECT DISTINCT Id, Amount FROM SalesPerson ORDER BY Amount ASC;

Based on the output of the above query, identify the correct statement.

- ☒ 1005 will be 3rd record
- ☐ 1009 will be 4th record
- ☐ 1002 will be 1st record
- ☐ 1001 will be 4th record

Q8 of 20

Consider the tables vehtype and vehicle given below:

vehtype(vid ,vtype) with vid being the primary key.

vehicle(id, vid, brand, model, price) with id being the primary key and vid foreign key to the vehtype table.

Consider the below join query:

```
select brand from vehicle v join vehtype vt
on v.vid=vt.vid
group by brand
having count(vtype)>1
```

Choose an equivalent subquery that would achieve the functionality performed by the above join query.

Note: The only difference between the options is in the 'WHERE' clause.

- ☐ SELECT brand FROM vehicle WHERE vid in (SELECT vid FROM vehtype HAVING COUNT(vtype)>1);
- ☐ SELECT brand FROM vehicle WHERE vid in (SELECT vid FROM vehtype GROUP BY vid HAVING COUNT(vtype)>1)
- ☐ SELECT brand FROM vehicle WHERE vid IN (SELECT vid FROM vehtype) HAVING COUNT(vid)>1;
- ☒ SELECT brand FROM vehicle WHERE vid in (SELECT vid FROM vehtype) GROUP BY brand HAVING COUNT(vid)>1

Table :emp

empno	ename	deptno	sal
1001	Mary	10	1000
1002	John	20	2000
1003	Jack	30	4000
1005	Jane	10	4000

empno is primary key

deptno is foreign key

Table:dept

deptno	dname
10	Finance
20	Accounts
30	Sales
40	Infrastructure

deptno is the primary key

Which of the following queries will execute successfully?

- ☒ INSERT INTO emp VALUES(1006, 'Fedrick', 10, 2000);
- ☒ INSERT INTO emp VALUES (1008, 'Fedrick', NULL, 3000);
- ☐ INSERT INTO dept (deptno) VALUES (40);
- ☐ INSERT INTO dept VALUES (NULL, 'HR');

Q10 of 20

Consider the table toys given below:

TOYID	TOYNAME	CATEGORYID	PRICE
200	Barbie	101	200
201	Drummer	102	500
202	Racing Car	103	600
203	Kelly	101	500
204	Ken Beat	102	400
205	Teddy	101	600
206	Monster Car	103	500

What will be the output of the below query?

SELECT * FROM toys t1 WHERE price = (SELECT MAX(price) FROM toys t2 WHERE t1.categoryid=t2.categoryid);

TOYID	TOYNAME	CATEGORYID	PRICE
201	Drummer	102	500
202	Racing Car	103	600
205	Teddy	101	600

A)

TOYID	TOYNAME	CATEGORYID	PRICE
200	Barbie	101	200
204	Ken Beat	102	400
206	Monster Car	103	500

B)

TOYID	TOYNAME	CATEGORYID	PRICE
204	Ken Beat	102	400
206	Monster Car	103	500

C)

TOYID	TOYNAME	CATEGORYID	PRICE
202	Racing Car	103	600
205	Teddy	101	600

D)

☒ A

☐ B

☐ C

☐ D

Q11 of 20

Cray Retail is a retail chain. They have a table in their database that has the following data:

EmployeeNo	EmployeeName	Department	Salary
1111	Paul	Sales	1000
1222	Smith	Sales	NULL
2111	Gary	Marketing	1000
2222	Manya	Marketing	1000
3111	Raj	HR	1000
3222	Paul	HR	NULL

Three developers Tom, Dick and Harry are given the task of finding the average salary of the employees in the company. They have been told that a NULL in the salary column means that those employees should not be considered.

They write the following queries:

Tom: SELECT SUM(Salary)/COUNT(*) FROM EMPLOYEE WHERE Salary IS NOT NULL;

Dick: SELECT SUM(Salary)/COUNT(*) FROM EMPLOYEE;

Harry: SELECT AVG(Salary) FROM EMPLOYEE;

Which of them have got the query correct?

☒ Tom and Harry have got it right

☐ Only Harry has got it right

☐ Tom and Dick have got it right

☐ Only Dick has got it right

Q12 of 20

Consider the table broker given below:

BrokerNo	Comission
101	30
102	20
103	NULL
102	20
101	40

Which of the following will be one of the rows in the output of the below SQL query?

```
SELECT BrokerNo, COUNT(NVL(Comission,0)) Commission  
FROM Broker GROUP BY BrokerNo;
```

A)

BrokerNo	Commission
103	0

B)

BrokerNo	Commission
102	3

C)

BrokerNo	Commission
103	1

D)

BrokerNo	Commission
103	null

☐ A

☐ B

☒ C

☐ D

Q13 of 20

Consider the below table named Destination:

DestId	DestName
101	Delhi
102	Pune
103	Chennai
104	Mumbai

DestId is the PRIMARY KEY.

Another table Travel holds the list of tour packages as given below:

TravelId	TourName	DestId
5001	Delhi Delight	101
5002	Capital Delight	101
5003	Mumbai Mazzaa	104
5004	Chennai Express	103
5005	Southern Spice	103

DestId in the Travel table references the DestId of the Destination table using a FOREIGN KEY.

Given the above details which of the following queries will execute successfully.

Choose 2 CORRECT options.

☒ DELETE FROM Destination where destId = 102

☐ UPDATE Destination SET destId = 105 where destId = 103

☐ DELETE FROM Destination where destId = 104

☒ UPDATE Destination SET destId = 105 where destId = 102

Q14 of 20

Consider the table Teacher given below:

TeacherId	FirstName	Location	Salary	Availability
T300	James	Chennai	5000	Y
T301	Klem	Mysore	5700	Y
T302	Nancy	Delhi	5000	Y
T303	Donald	Pune	6500	N
T304	Ervin	Delhi	4750	N

How many rows will get updated when the below query is executed?

UPDATE TEACHER SET SALARY = SALARY + 5000 WHERE TEACHERID IN

(SELECT TEACHERID FROM TEACHER WHERE AVAILABILITY = 'N' OR SALARY > 5000);

☐ 1

☐ 4

☒ 3

☐ 2

Q15 of 20

Consider the table products given below:

SQL> SELECT * FROM products;

MANUFACTURER	MODELNO	PRODTYPE
ABC	1000	PC
ABC	1001	PC
ABC	1002	Printer
ABC	1003	Laptop
ABC	1004	Printer
ABC	1005	Printer
ABC	1006	Laptop
DEF	2000	PC
DEF	2001	Laptop
GHI	3000	Laptop
FGH	4000	PC
FGH	4001	Printer
FGH	4002	Laptop

Note: modelno is the PRIMARY KEY of products table.

What will be the output of the following query?

```
SELECT prodtype FROM products
GROUP BY prodtype
HAVING COUNT(modelno) = (SELECT MIN(COUNT(modelno)) FROM products GROUP BY prodtype);
```

- ☐ PC
- ☒ PC
 Printer
- ☐ Laptop
- ☐ Printer

Q16 of 20

Consider the tables SUPPLIER and ORDERS given below:

TABLE: ORDERS

ORDER_ID	SUPPLIER_ID	ORDER_DATE
88891	10001	20-Jun-15
88891	10002	23-Feb-15

TABLE: SUPPLIER

SUPPLIER_ID	SUPPLIER_NAME
10001	MICROSOFT
10002	HONEYWELL
10003	NOKIA
10004	SAMSUNG

Identify the query to fetch the details of all the suppliers along with order details. Include the suppliers who haven't ordered any items also.

- ☐ SELECT s.supplier_id, o.supplier_id, o.order_date FROM supplier s RIGHT OUTER JOIN orders o on s.supplier_id=o.supplier_id;
- ☐ SELECT s.supplier_id, o.supplier_id, o.order_date FROM orders o LEFT OUTER JOIN supplier s on s.supplier_id=o.supplier_id;
- ☐ SELECT s.supplier_id, o.supplier_id, o.order_date FROM orders o FULL OUTER JOIN supplier s on s.supplier_id=o.supplier_id;
- ☒ SELECT s.supplier_id, o.supplier_id, o.order_date FROM supplier s LEFT OUTER JOIN orders o on s.supplier_id=o.supplier_id;

Q17 of 20

Consider the tables customer and subscription given below:

Table: customer

customerId	customerName
101	Jack
102	Tom
103	Harry
104	Peter

Table: subscription

subscriptionId	customerId	channelId
501	101	201
502	101	202
503	102	203
504	102	201
505	103	201

What is the output of the below query?

```
SELECT customerName
FROM customer c JOIN subscription s ON s.customerId=c.customerId
WHERE s. customerId NOT IN (SELECT customerId
FROM subscription GROUP BY customerId HAVING COUNT (customerId) =
(SELECT MAX(COUNT(customerId)) FROM subscription GROUP BY customerId));
```

- ☐ Jack
- ☒ Harry
- ☐ Tom
- ☐ Peter

Table:Bank

BankCode	BankName	Location
B101	ICICI	New Delhi
B102	HSBC	Bangalore
B103	HDFC	Chennai
B104	SBI	Mumbai

Table: CustAccountDetails

CustAccId	BankCode	AccId	Balance
CA01	B101	AC101	500000
CA02	B102	AC102	550000
CA03	B101	AC101	600000
CA04	B103	AC103	700000
CA05	B104	AC104	400000

Table: AccountDetails

AccId	AccType
AC101	Saving
AC102	Demat
AC103	Current
AC104	Recurring

What would be the output of below query?

```
SELECT CustAccId, Balance, BankName, ad.AccId FROM CustAccountDetails cd
INNER JOIN Bank ba ON ba.BankCode = cd.BankCode INNER JOIN AccountDetails ad
ON cd.AccId = ad.AccId
WHERE Balance >
(SELECT AVG(Balance) FROM CustAccountDetails
INNER JOIN Bank
ON Bank.BankCode = CustAccountDetails.BankCode
WHERE BankName = ba.BankName) AND ad.AccType = 'Saving';
```

CustAccId	Balance	BankName	AccId
CA03	600000	ICICI	AC101
CA01	500000	ICICI	AC101

A)

CustAccId	Balance	BankName	AccId
CA04	700000	HDFC	AC103
CA03	600000	ICICI	AC101

B)

CustAccId	Balance	BankName	AccId
CA03	600000	ICICI	AC101

C)

D) No rows will be selected

☐ A

☐ B

☒ C

Table:Bank

BankCode	BankName	Location
B101	ICICI	New Delhi
B102	HSBC	Bangalore
B103	HDFC	Chennai
B104	SBI	Mumbai

Table: CustAccountDetails

CustAccId	BankCode	AccId	Balance
CA01	B101	AC101	500000
CA02	B102	AC102	550000
CA03	B101	AC101	600000
CA04	B103	AC103	700000
CA05	B104	AC104	400000

Table: AccountDetails

AccId	AccType
AC101	Saving
AC102	Demat
AC103	Current
AC104	Recurring

What will be the output of the following query?

```
SELECT B.BankName,AD.AccType,SUM(Balance)
FROM CustAccountDetails CA INNER JOIN Bank B ON CA.BankCode = B.BankCode
INNER JOIN AccountDetails AD ON CA.AccId = AD.AccId
GROUP BY B.BankName,AD.AccType HAVING SUM(Balance) =
(SELECT MIN(SUM(Balance)) from custAccountDetails GROUP BY BankCode,AccId);
```

BankName	AccType	Sum(Balance)
HSBC	Demat	550000
HDFC	Current	700000
SBI	Recurring	400000

A)

BankName	AccType	Sum(Balance)
SBI	Recurring	400000

B)

BankName	AccType	Sum(Balance)
SBI	Recurring	400000
ICICI	Saving	1100000

C)

BankName	AccType	Sum(Balance)
ICICI	Saving	1100000

D)

☐ A

☒ B

Consider following tables:

Table: Donor

DONORID	DONORNAME	BLOODGROUP
D101	Flor	O+
D102	Raul	B-
D103	Sam	AB+
D104	Robert	B+

Table: BloodTransaction

TRANSID	DONORID	PATIENTID	TRANSDATE
T101	D104	P105	01-JUN-13
T102	D101	P102	19-AUG-13
T103	D104	P104	20-SEP-13
T104	D101	P102	28-NOV-13
T105	D102	P101	12-MAR-14

There is a requirement to display donor id, donor name of those donors who donated the blood.

Also display the patient id who have received blood from these donors.

The following query was written to solve the above requirement.

```
SELECT DISTINCT d.donorid, donorname, patientid FROM donor d INNER JOIN bloodtransaction b ON d.donorid = b.donorid;
```

What is the output of the above query?

DONORID	DONORNAME	PATIENTID
D104	Robert	P105
D101	Flor	P102
D104	Robert	P104
D101	Flor	P102
D102	Raul	P101

A)

DONORID	DONORNAME	PATIENTID
D104	Robert	P104
D102	Raul	P101
D101	Flor	P102
D104	Robert	P105

B)

C) Query written is correct but fetches no rows

D) Query written does not satisfy the stated requirement

☐ A

☒ B

☐ C